Title: A Multi-Modal Ensemble Learning Approach for Early Survival Prediction

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Background

Early survival prediction (≤6 vs. >6 months) in lung cancer patients is critical for optimizing treatment strategies. Modern diagnostics generate multi-modal data (clinical, molecular, imaging), yet missing modalities in real-world settings challenge traditional machine learning approaches.

Methods

We propose an ensemble method for multi-modal classification that naturally handles missing data without imputation. Validated on Apollo11, an Italian lung cancer dataset, our approach trains individual classifiers for each modality: real-world clinical data (RWD), radiomics (CT scans), digital pathology, and Fluorescence-activated Cell Sorting (FACS). Predictions are combined via weighted sums reflecting modality importance, avoiding patient exclusion due to incomplete data.

Results

Among 664 patients, all had RWD, while 373 included radiomics, 330 digital pathology, and 155 FACS; only 42 had all modalities. The ensemble's predictive performance did not show significant improvement over multimodal models. However, it remained robust even when modalities were removed, suggesting that the model can leverage the available data despite the limited dataset and the scarcity of patients with all modalities.

Conclusion

The results demonstrate that our approach is robust to missing modalities and provides competitive performance compared to traditional methods, while offering transparency, as it is built on simple models, and ensuring flexibility in integrating heterogeneous data sources from clinical practice and advanced diagnostic techniques. The next step we envision is the application of such models using a significantly larger amount of patients having all the modalities to study in-depth the information that are also relevant in terms of model's performance improvement.